Mechanical waves and distant effects

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use travel-time reasoning while distinguishing waves from the motion of material.

SUCCESS CRITERIA - I can...

1. Identify the medium carrying the wave.
2. Use speed and distance to estimate travel time.
3. Explain why a simple travel time does not predict damage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A converging lens has f=15 cm and d_o=6 cm. Use $1/f=1/d_o+1/d_i$ and $m=-d_i/d_o$. Find d_i, magnification and image kind.

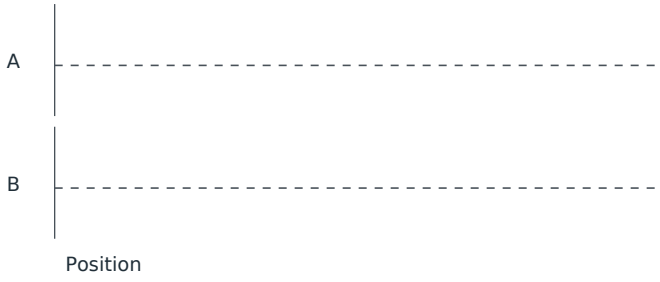
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: d = v Δt; Δt = d/v.
Use the supplied wave speed for its medium and wave type.
These classroom models are not real hazard forecasts.

Worked reasoning: In a simple model, a wave amplitude falls from 4 to 2 units while intensity is proportional to amplitude squared. Intensity becomes one quarter, not one half.

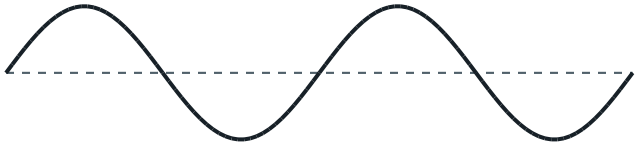
Your turn: Draw the two waves and two corresponding relative-intensity bars.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Draw a disturbance traveling through water and mark local water motion. Why does wave arrival not mean the original water crossed the whole ocean?

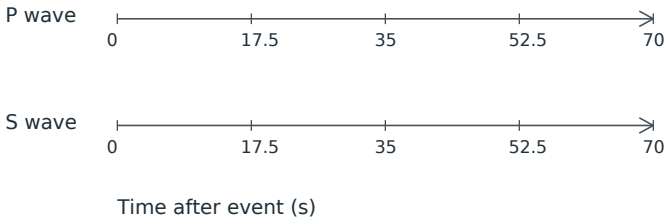


Sketch and label your response.

Marzano check - circle: ① ② ③

04 Apply the physics

A simplified seismic model uses P-wave speed 6 km/s and S-wave speed 3.5 km/s. A station is 210 km from the source. Calculate each arrival time and the S-P delay. Mark both arrivals on the timeline.



Marzano check - circle: ① ② ③

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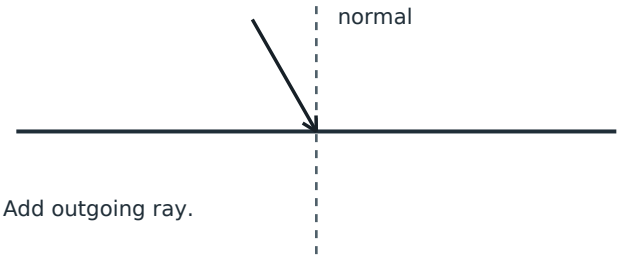
05 Explain your reasoning

Why does knowing wave arrival time alone not establish local damage? Name other evidence.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Light moves from a transparent medium with n=1.7 into air (n=1). Find critical angle using sin θ_c=1/n. Describe incidence above it in the ideal model.
Label the media, normal and ray directions.



Add outgoing ray.

Marzano check - circle: 1 2 3

07 Exit check

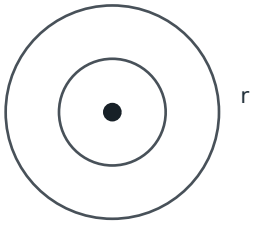
A shallow-water model uses v=√(gh). With g=10 and h=100 m, find speed and time for 10 km. State a limitation.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A signal has intensity 8 W/m² at 10 m from an ideal point source. Find intensity at 30 m and total radiated power. Explain one limitation for ground waves.



Label distances and compare intensity.

Marzano check - circle: 1 2 3